

Clinical Motivation

Early atrophy (EA) represents an important precursor to more advanced atrophic changes in age-related macular degeneration (AMD). In baseline optical coherence tomography (OCT) scans, clinicians have additionally observed a distinctive vertical hypertransmission pattern, informally described as “**barcoding**,” which they hypothesize may identify regions at increased risk of subsequent progression.

Unlike established OCT biomarkers, barcoding does not yet have a standardized quantitative definition. Assessment is primarily visual and difficult to reproduce at scale.

Objective

Translate these observations into measurable imaging features by automatically localizing and quantifying **EA and barcoding at baseline**. The long-term goal is to determine whether their presence, extent, and spatial characteristics improve prediction of atrophic progression.

Scan Preparation

1. Load a selected B-scan or volume and its segmentation boundaries from the HEYEX E2E volume.
2. Flatten the scan to BM and retain 150 pixels below it.
3. Standardize brightness using a z-score across the whole ROI
4. Apply mild Gaussian denoising (1.0, 0.5) while preserving vertical patterns.

Methodology

Each feature $F_j(x)$ is placed on a scan-relative scale using its median and interquartile range (IQR) along with a small constant $\epsilon = 10^{-6}$ to avoid division by zero. The calibrated feature vector contains median and 90% intensity, depth continuity, and verticality.

Depth continuity. Within a local window, horizontal intensity patterns are correlated between rows separated by the selected depth lag $d = 4$ pixels:

$$C(x) = \frac{1}{D-d} \sum_{y=1}^{D-d} \frac{(\mathbf{l}_{y,x} - \bar{\mathbf{l}}_{y,x})^T (\mathbf{l}_{y+d,x} - \bar{\mathbf{l}}_{y+d,x})}{\|\mathbf{l}_{y,x} - \bar{\mathbf{l}}_{y,x}\|_2 \|\mathbf{l}_{y+d,x} - \bar{\mathbf{l}}_{y+d,x}\|_2 + \epsilon}.$$

Verticality. Smoothed horizontal and depth gradients form the local structure tensor $\mathbf{J}$:

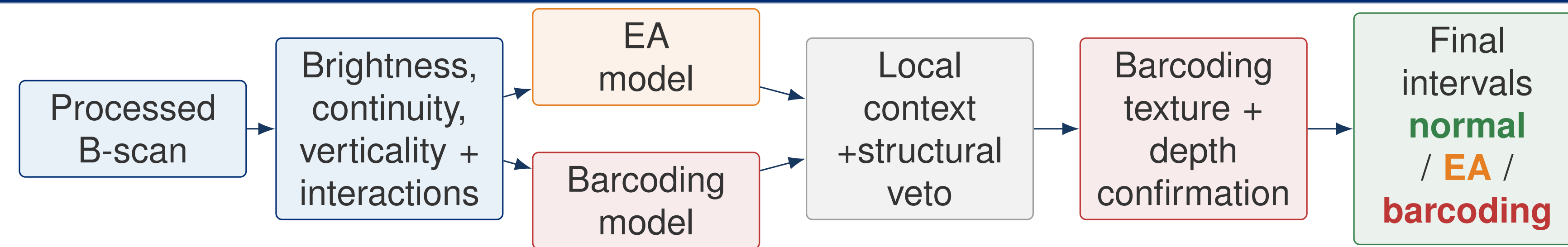
$$\mathbf{J} = G_{\sigma_t} * \begin{bmatrix} I_x^2 & I_x I_z \\ I_x I_z & I_z^2 \end{bmatrix}, \quad V = \frac{\lambda_1 - \lambda_2}{\lambda_1 + \lambda_2 + \epsilon} \cos^2 \theta,$$

where $\lambda_1 \geq \lambda_2$ are the tensor eigenvalues and θ is its dominant gradient orientation. V approaches one for a strongly oriented vertical pattern; the model uses the strong-vertical pixel fraction within each column.

Separate logistic models convert these measurements into probabilities:

$$p_k(x) = \frac{1}{1 + \exp[-(\beta_{0k} + \beta_k^T \mathbf{z}(x))]}, \quad k \in \{\text{EA, barcode}\}.$$

Model Framework



EA and barcoding are scored separately because they have different visual criteria. Structural and vessel-like regions are vetoed before final intervals are formed. The selected configuration applies Gabor texture and near-depth confirmation only to barcoding.

Current Calibrated Specification.

Barcoding

- ▶ At least 55% local support within ± 5 columns
- ▶ Interval mean/peak probability $\geq 0.52/0.68$
- ▶ At least two feature votes
- ▶ Minimum width 8; maximum bridged gap 4
- ▶ Gabor texture and near-depth confirmation

Early Atrophy (EA)

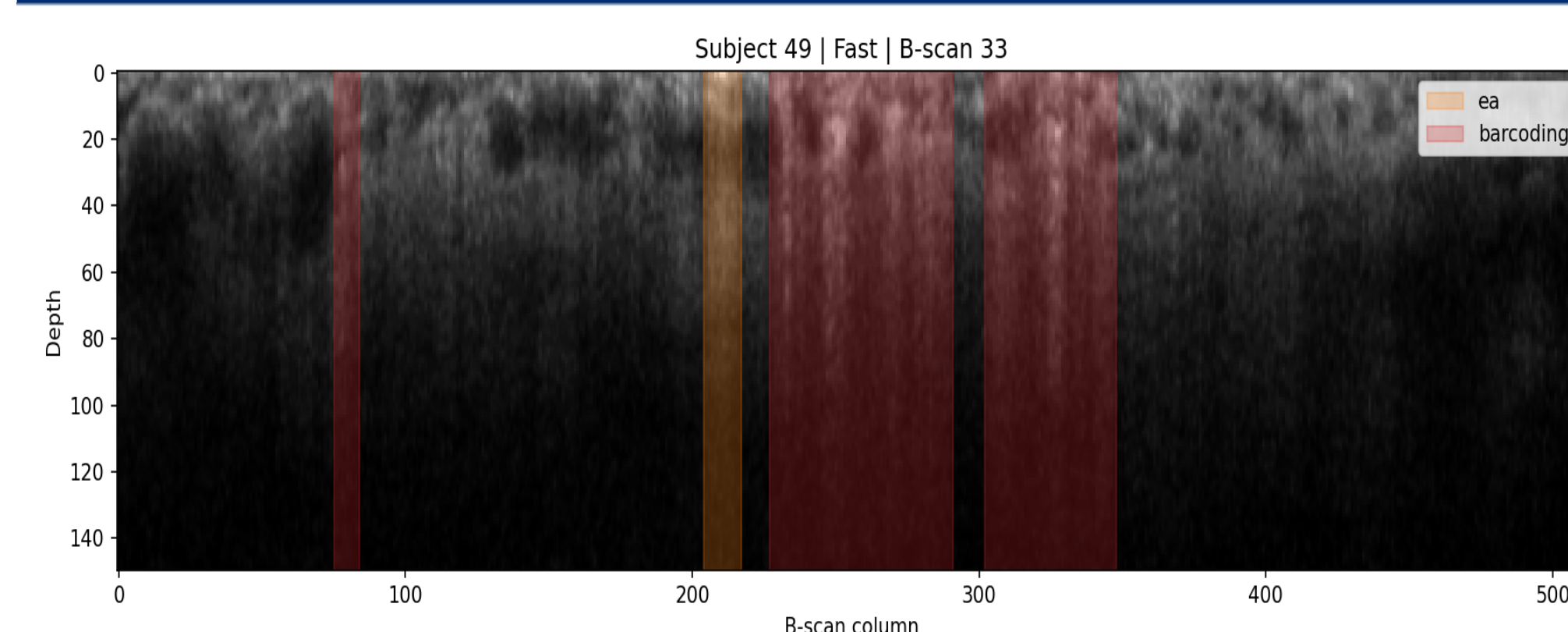
- ▶ At least 60% local support within ± 7 columns
- ▶ Interval mean/peak probability $\geq 0.48/0.62$
- ▶ At least three feature votes
- ▶ Minimum width 12; maximum bridged gap 5
- ▶ No Gabor-texture or depth-band gate

A candidate is retained when it demonstrates sufficiently strong vertical-pattern energy together with increased brightness near Bruch’s membrane. Hypertransmission is evaluated across near (0–20%), middle (20–50%), and deep (50–100%) sub-BM regions to characterize how the signal persists with depth. A separate structural model and dark-shadow criterion exclude regions more consistent with vessels, anatomical structures, or shadowing than true barcoding.

Scale-up tuning algorithm.

1. **Patient-grouped validation.** Use nested cross-validation; tune in inner folds and evaluate only in untouched outer folds.
2. **Model calibration.** Refit the L2-regularized logistic models with the L-BFGS optimizer.
3. **Threshold optimization.** Use Optuna TPE Bayesian optimization to maximize patient-averaged Dice while penalizing false-positive intervals.
4. **Lock and test.** Aggregate inner-fold settings, freeze one configuration, and evaluate it on an independent cohort. Uncertain labels are excluded; vessel/structural labels are explicit negatives. TPE jointly tunes probabilities and thresholds.

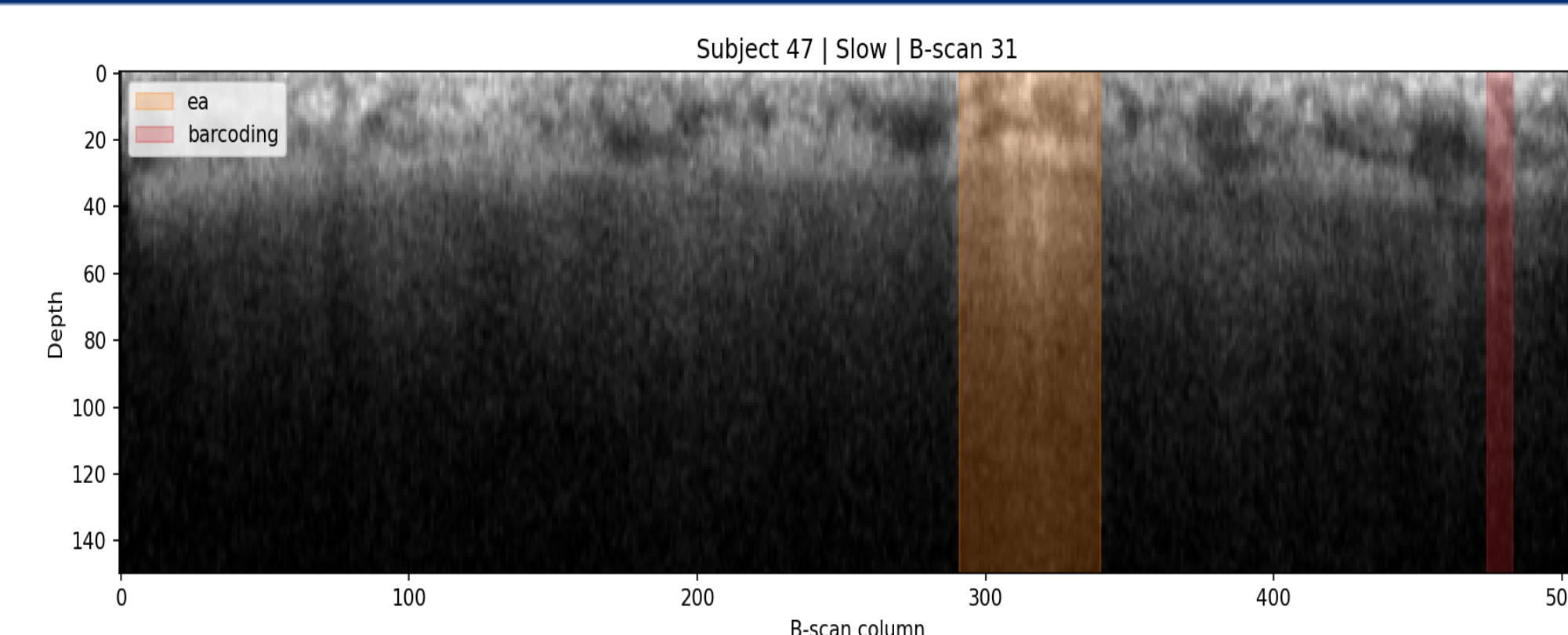
Results



Fast Barcoding Progressor

Precision 0.92; recall 0.71; Dice 0.81

Consecutive positive columns were grouped into EA or barcoding intervals. We measured presence, number of intervals, location, width, and total horizontal burden for each phenotype. Total burden represents the proportion of the B-scan width affected by EA or barcoding and does not represent physical area or volume.



Slow Barcoding Progressor with EA

Precision 1.00; recall 0.91; Dice 0.95

Development-Set Performance

Ten selected B-scans were manually annotated: five fast-progressing and five slow-progressing cases. Uncertain and vessel/structural regions were excluded from scoring.

Finding	Precision	Recall	Dice
Barcoding, selected	0.817	0.520	0.636
EA, selected combined	0.462	0.699	0.557
EA, best EA-only V3	0.475	0.699	0.566

Precision fewer false alarms	Recall more disease recovered	Dice overall overlap
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Statistical Analysis

Example model comparison. We specify two nested time-to-progression models to test whether the automatically quantified barcoding biomarker adds information beyond standard baseline predictors.

Model 1: Baseline Covariates Only

$$h_i^{(1)}(t) = h_0(t) \exp(\gamma^T \mathbf{Z}_i),$$

where $\mathbf{Z}_i$ includes established OCT biomarkers, and using robust standard errors

Model 2: Baseline Covariates and Barcoding Biomarker

$$h_i^{(2)}(t) = h_0(t) \exp[\gamma^T \mathbf{Z}_i + \beta_N N_i + \beta_w \bar{w}_i + \beta_{Nw}(N_i \bar{w}_i)].$$

Here N_i is the number of baseline barcoding intervals, each detected width $w_{ir} = e_{ir} - s_{ir} + 1$ is retained, and $\bar{w}_i$ is the mean interval width used in the primary fixed-length model. The interaction $N_i \bar{w}_i$ tests whether interval width has a different association with risk when barcoding is more fragmented or numerous. Total and maximum width can replace $\bar{w}_i$ in sensitivity analyses.

Comparison. After adjusting for baseline covariates, barcoding was associated with faster atrophic progression (adjusted HR = 1.42, 95% CI: 1.15–1.76). Adding barcoding to the clinical/EA-only model significantly improved model fit (likelihood-ratio p=0.008) and increased cross-validated concordance from 0.71 to 0.76. Calibration remained good, with predicted progression risk closely matching observed risk across follow-up. These results would suggest that baseline barcoding provides independent prognostic information beyond EA alone.

Patient-level data splitting is required so scans from the same patient do not appear in both training and validation. This comparison estimates the **incremental prognostic value** of barcoding rather than merely its correlation with existing baseline severity. Proportional hazards are assessed with scaled Schoenfeld residuals. If both eyes are included, uncertainty will be assessed with patient-clustered robust standard errors (or a patient frailty sensitivity model).